Activity: Factory Floor Layout Design

Prepared for the assignment based on the three given industrial scenarios.

Scenario	Most Suitable Layout	Reason
1. High-Volume Car Manufacturing	Product / Line Layout	The work follows one fixed sequence with continuous flo
2. Multi-Process Engineering Workshop	Process / Functional Layout	Different jobs need different routes, so machines must be
3. Electronics Production Units	Cellular Layout	Machines are arranged in small cells for families of simila

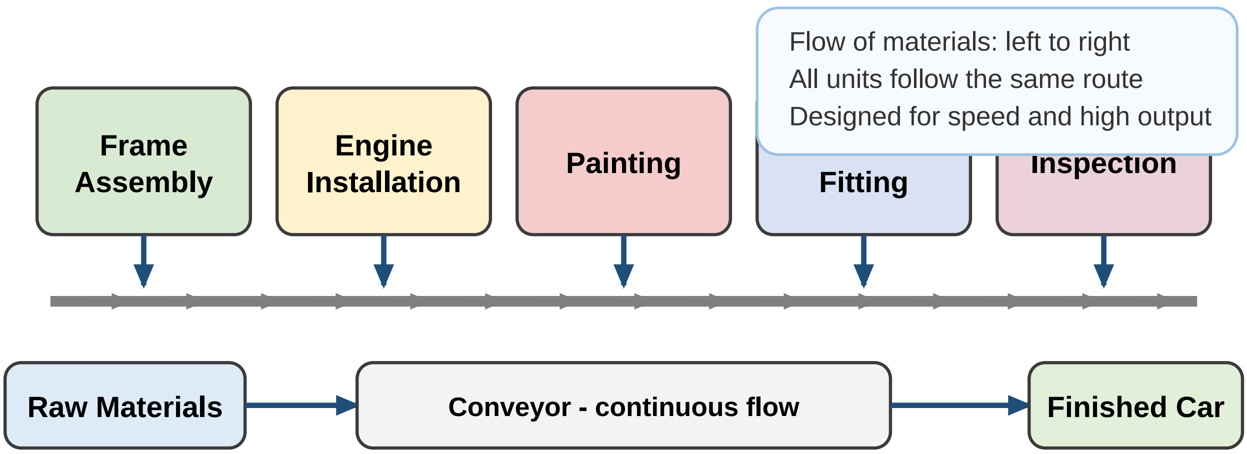
Included in each scenario page: identified layout type, digital floor layout, material flow arrows, machine/workstation labels, and a short explanation.

Scenario 1 - High-Volume Car Manufacturing

Selected layout: Product Layout (Line Layout)

This layout is suitable because the same car model is produced in very large quantities and every unit follows the same sequence of operations. Machines and workstations are arranged in the exact production order so the conveyor can move cars continuously from one stage to the next.

Digital Floor Layout - Product / Line Layout



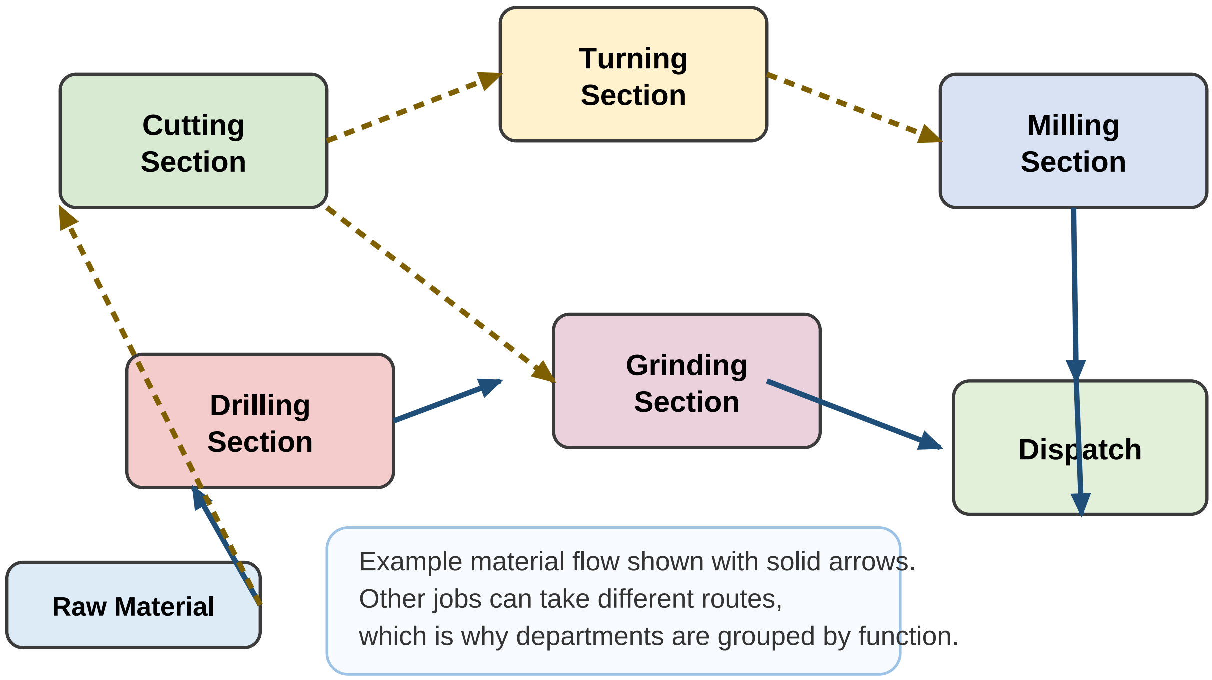
Short explanation: The workstations are placed in one straight sequence because production is repetitive and standardized. Material handling is simple because a conveyor moves each car from one stage to the next. This reduces waiting time, increases efficiency, and supports high-volume output.

Scenario 2 - Multi-Process Engineering Workshop

Selected layout: Process Layout (Functional Layout)

This workshop makes different custom parts, so each order may require a different operation sequence. Therefore, similar machines are grouped together by function, and material moves between departments according to the needs of each job.

Digital Floor Layout - Process / Functional Layout

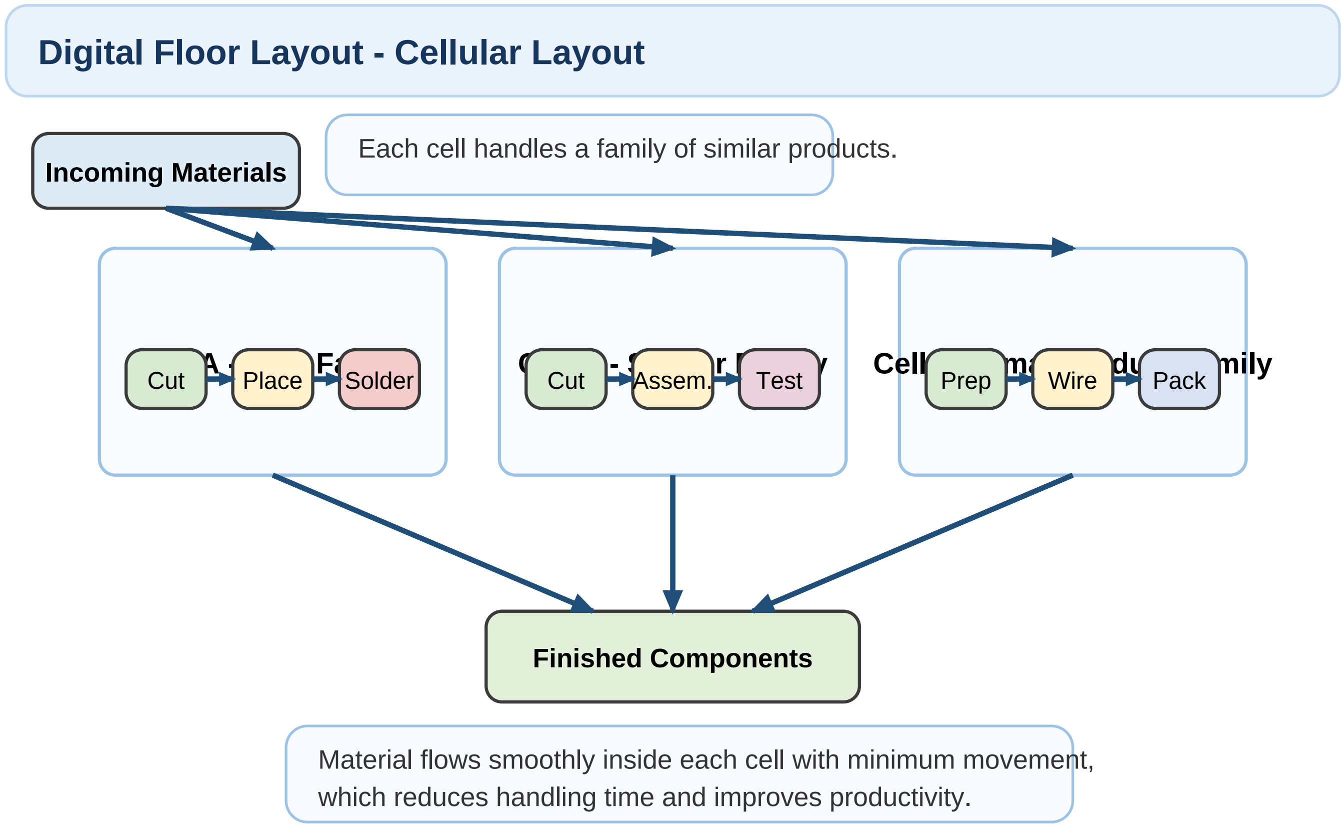


Short explanation: A process layout is best because the company handles many different custom orders. Machines are grouped into functional departments such as cutting, drilling, and milling, so any part can visit only the sections it needs. This gives the workshop high flexibility and adaptability.

Scenario 3 - Electronics Production Units

Selected layout: Cellular Layout

The factory produces different electronic components, but many items share similar manufacturing steps. Grouping machines into small cells for similar product families reduces travel distance, speeds up flow, and improves productivity.



Short explanation: A cellular layout is most suitable because similar electronic products can be processed in dedicated machine groups. Each cell is arranged in logical sequence, so material travels a shorter distance and flows smoothly. This lowers handling time and increases productivity.